

Integrated Circuits and Optoelectronics

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1 Section 22 — Integrated Circuits

Definition

An **integrated circuit (IC)** is a miniaturized circuit that contains a number of resistors, capacitors, diodes and transistors stuffed together on a single chip of silicon no bigger than your fingernail. The component count on one chip may vary from just a few to millions.

1.1 Inside an IC

The trick to cramming everything into such a small package is to make all the components out of tiny *n*-type and *p*-type silicon structures that are embedded into the silicon chip during the production phase — exactly the doped structures met in the semiconductor and transistor sections. To connect the little transistors, resistors, capacitors and diodes together, aluminium plating is applied along the surface of the chip; the plating is the “wiring”.

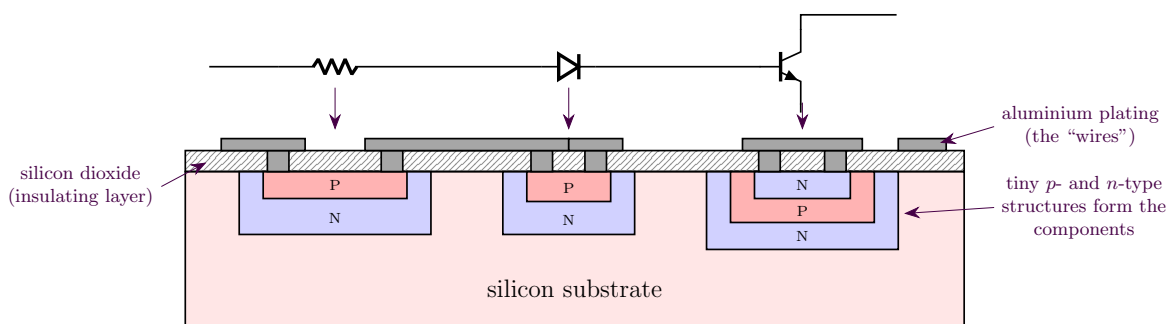


Figure 1: Magnified cross-section of an IC. Each circuit component is a small arrangement of *p*- and *n*-type structures embedded in the substrate; a silicon-dioxide layer insulates the surface, and aluminium plating on top links the components together.

tiny <i>p/n</i> structures	→ components	aluminium plating	→ wires	one
chip → a whole circuit				

1.2 Kinds of ICs

ICs come in analog, digital, or analog/digital form:

- **Analog** (or linear) ICs produce, amplify, or respond to *varying* voltages. Common examples: voltage regulators, operational amplifiers, comparators, timers and oscillators.
- **Digital** (or logic) ICs respond to or produce signals having only *high and low* voltage states. Common examples: logic gates (AND, OR, NOR, ...), microcontrollers, memories, counters, shift registers, multiplexers, encoders and decoders.
- **Analog/digital** ICs share properties of both — e.g. an analog timer containing a digital counter, or an IC that reads digital information and produces a linear output to drive a stepper motor or an LED display.

ICs are so pervasive that you are likely to use them in any project you will undertake.

1.3 IC Packages

ICs come in many and various packages. The determining factors are the *number of pins* and the *power dissipation* — a high-power voltage-regulator IC may have three pins and look just like a high-power transistor. The majority of ICs, however, have many more pins and come in a **dual in-line (DIL/DIP)** package of 8, 14, 16, 20, 24 or 40 pins, or in one of its surface-mount shrinkages.

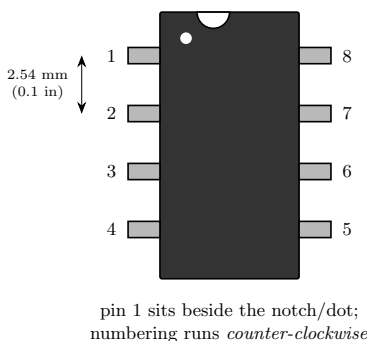


Figure 2: The dual in-line package (here 8 pins, top view). Pin 1 sits beside the notch/dot and the numbering runs counter-clockwise around the package.

Table 1: Common IC packages.

Package	Long name	Pitch (mm)	Notes
DIL (DIP)	Dual in-line	2.54	the breadboard-friendly one surface mount
SO/SOIC/SOP	Small-outline IC package	1.27	
MSOP/SSOP	Mini/shrink small outline	0.65	
SOT	Small-outline transistor	0.65	
TQFP	Thin quad flat pack	0.8	pins on all four sides
TQFN	Thin quad flat, no leads	0.4–0.65	pads underneath the body

The same IC is often available in multiple packages, making it possible to prototype in something easy to solder (DIL or SO) and then switch to a smaller package for the final product. The finest-pitch packages put two pins in every millimetre — not really intended for hand soldering.

1.4 The CMOS Inverter — the IC Building Block

What actually fills a modern digital IC? Overwhelmingly, **complementary MOSFET (CMOS)** logic. The transistor notes built logic gates from NMOS switches and a pull-up *resistor*; CMOS replaces that resistor with a PFET, giving a complementary pair:

The complementary pair

An **NFET** conducts when its gate is *high*; a **PFET** conducts when its gate is *low*. Tie both gates to the same input and exactly *one* of the two is ON at any time:

$V_{in} = V_{DD}$: NFET ON, PFET OFF

$V_{in} = 0$: PFET ON, NFET OFF

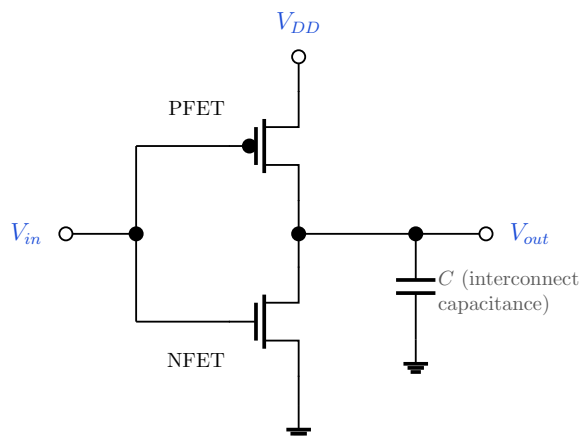


Figure 3: The CMOS inverter. The PFET connects the output to V_{DD} ; the NFET connects it to ground; the input drives both gates. C is the capacitance of whatever the output drives (interconnect, the next gate's inputs).

1.4.1 Example 1 — States of the CMOS Inverter

Using the S model for both transistors, determine V_{out} of the CMOS inverter of Figure 3 for $V_{in} = V_{DD}$ and for $V_{in} = 0$. *Verify that no current flows from V_{DD} to ground in either state.*

Solve: Take each input level in turn and replace each FET by an open or closed switch:

- $V_{in} = V_{DD}$ — the NFET sees a high gate and closes; the PFET sees a high gate and opens. The output is connected to ground through the NFET: $V_{out} = 0$.
- $V_{in} = 0$ — the PFET sees a low gate and closes; the NFET opens. The output is connected to the supply through the PFET: $V_{out} = V_{DD}$.

Table 2: The inverter's truth table.

V_{in}	PFET	NFET	V_{out}	supply-to-ground path?
V_{DD}	OFF	ON	0	no — PFET blocks
0	ON	OFF	V_{DD}	no — NFET blocks

Verify: In *both* states one of the two switches is open, so the path from V_{DD} to ground is always broken — the gate consumes essentially no power while it sits in either state. Current flows only during the brief moment of switching, to charge or discharge the load capacitance C .

Why CMOS conquered the IC

No resistor, no static current, no static power. A pull-up-resistor gate burns current the whole time its switch is closed; a CMOS gate burns power only when it *switches*. Multiply that saving by millions of gates on one chip and CMOS is the only game in town.

NFET pulls down || PFET pulls up → exactly one ON → rail-to-rail output,
zero static current

2 Section 23 — Optoelectronics

Definition

Optoelectronics is the branch of electronics that deals with *light-emitting* and *light-detecting* devices. Light-emitting devices (lamps, light-emitting diodes) use an electric current to excite electrons into higher energy levels — when an electron falls back down, a photon is emitted. Light-detecting devices (photoresistors, photodiodes, phototransistors) do the reverse: they use incoming photons to liberate bound electrons within semiconductor materials, converting electromagnetic energy into currents and voltages.

Light emitters are used for illumination and as indicator lights; light detectors are used in light-sensing and communication circuits, such as dark-activated switches and remote controls.

emitters: current in → photons out detectors: photons in →
current out

2.1 A Little Lecture on Photons

Photons are the elemental units of electromagnetic radiation. White light is composed of many different kinds of photons — some blue, some red, and so on; there is no such thing as a *white* photon. Photons are not limited to visible light either: there are radio-frequency photons, infrared photons, microwave photons, X-ray photons — our eyes simply cannot detect them.

A photon has no rest mass, but it carries energy and momentum, and it has a distinct wavelike character. The trick to *making* a photon is to accelerate (or decelerate) a charged particle: electrons vibrating in an antenna make radio photons; outer-shell electrons jumping between atomic energy levels make visible photons; charge accelerations inside nuclei make gamma rays.

The two photon relations

$$\lambda = \frac{v}{f} \qquad E = hf = \frac{hc}{\lambda}$$

v is the photon's speed ($v = c = 3.0 \times 10^8$ m/s in free space, smaller in glass) and $h = 6.63 \times 10^{-34}$ J · s is Planck's constant. A handy shortcut when λ is in nanometres:

$$E \text{ (eV)} = \frac{1240}{\lambda \text{ (nm)}}$$

Long wavelength = low frequency = low energy; short wavelength = high frequency = high energy.

2.2 The Electromagnetic Spectrum

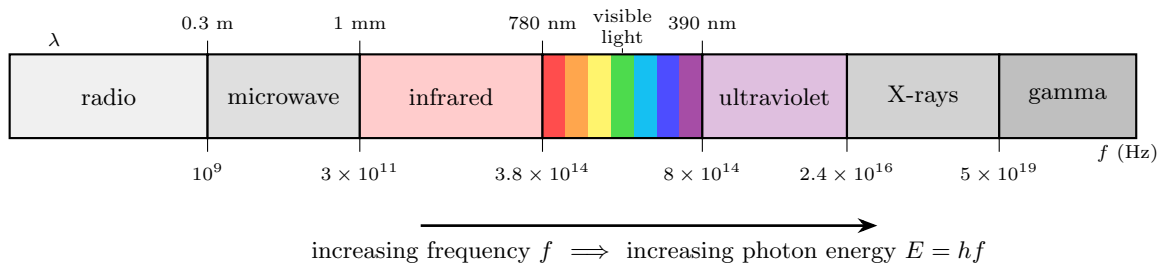


Figure 4: The electromagnetic spectrum (band widths not to scale). Visible light is the narrow slice from roughly 3.8×10^{14} to 7.7×10^{14} Hz — about 390 to 780 nm.

Table 3: The bands and where they come from.

Band	Frequency	Typical origin
Radio	up to 10^9 Hz	alternating currents in circuits and antennas
Microwave	10^9 – 3×10^{11} Hz	atomic transitions, electron/nuclear spins
Infrared	3×10^{11} – 3.8×10^{14} Hz	molecular oscillations, hot objects, LEDs
Visible	3.8 – 7.7×10^{14} Hz	outer-electron rearrangements in atoms
Ultraviolet	8×10^{14} – 3.4×10^{16} Hz	long electron jumps from highly excited states
X-rays	2.4×10^{16} – 5×10^{19} Hz	rapid deceleration of fast charged particles
Gamma rays	above 5×10^{19} Hz	transitions within atomic nuclei

2.2.1 Example 2 — The Energy of a Photon

Find the energy of (a) a red photon ($\lambda = 650$ nm) and (b) a blue photon ($\lambda = 470$ nm), in joules and in electron-volts.

Solve: Apply $E = hc/\lambda$:

$$E_{\text{red}} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{650 \times 10^{-9}} = 3.06 \times 10^{-19} \text{ J} = 1.91 \text{ eV}$$

$$E_{\text{blue}} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{470 \times 10^{-9}} = 4.23 \times 10^{-19} \text{ J} = 2.64 \text{ eV}$$

The shortcut gives the same at a glance: $1240/650 = 1.91$ eV and $1240/470 = 2.64$ eV.

Why this matters for the next section

A blue photon carries more energy than a red one, so a blue-emitting junction needs a *larger* band gap — and a diode with a larger band gap needs a *larger forward voltage* to conduct. This single fact explains the ordering of the LED curves in Figure 8: infrared turns on near 1.2 V, red near 1.8 V, blue and white not until 3.4–3.6 V.

3 Section 24 — Light-Emitting Diodes

Definition

Light-emitting diodes (LEDs) are two-lead devices similar to pn -junction diodes, except that they are designed to emit visible or infrared light. When the anode is made more positive than the cathode by at least the forward voltage (0.6 to 2.2 V for many types, more for blue), current flows and light is emitted. Reverse the polarity and the LED neither conducts nor emits.

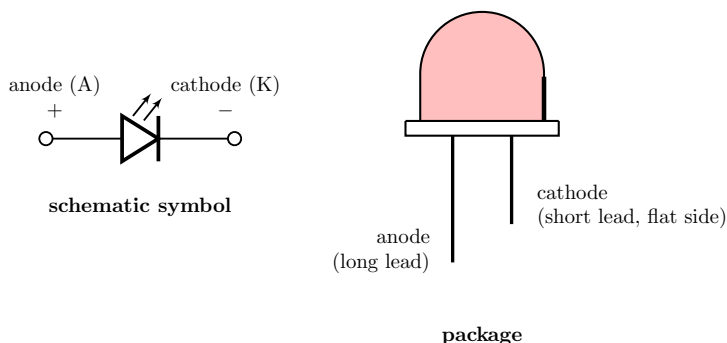


Figure 5: LED symbol and package. The two outward arrows mean *emits light*. On the package, the cathode is marked by the shorter lead and the flat side of the dome.

Historically red came first; yellow, green and infrared followed, and blue only arrived in the 1990s (the 2014 Physics Nobel Prize was awarded for it). Today LEDs come in essentially every colour including white, plus high-power illumination types and organic LEDs (OLEDs) for displays.

3.1 How an LED Works

The light-emitting section of an LED is a pn junction. Forward bias it, and electrons from the n side are pushed across the junction into the p side, where they combine with holes — and as they combine, photons are emitted. The junction sits in an epoxy shell doped with light-scattering particles; a reflector cup beneath the chip directs light upward, and heavy-gauge leads wick heat away.

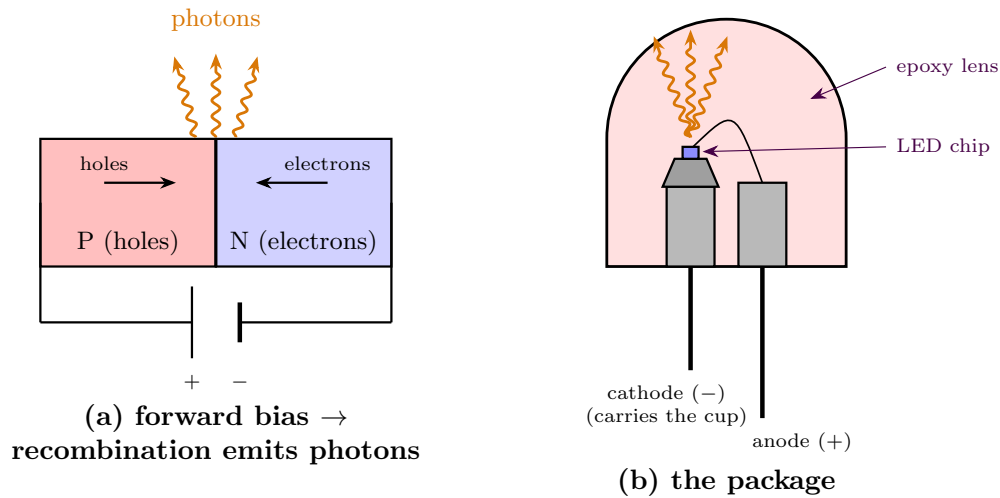


Figure 6: (a) Recombination at the forward-biased junction releases photons. (b) Inside the package: the chip sits in a reflector cup (on the cathode post), under an epoxy lens.

forward bias $\rightarrow$ electrons meet holes at the junction $\rightarrow$ each recombination emits one photon

Why LEDs are not made of silicon

Silicon and germanium have *indirect* band gaps: an electron cannot drop across the gap without help from a lattice vibration, so recombination mostly makes *heat*, not light. Compound semiconductors such as GaAs, GaN, InGaAlP and GaAsP have *direct* band gaps — the electron drops straight down and hands its energy to a photon. That is why every LED dye material in Table 4 is a III–V compound, and none is plain silicon.

3.2 Kinds of LEDs

- **Visible-light LEDs** — cheap, durable indicator lights. Green (~ 565 nm), yellow (~ 585 nm), orange (~ 615 nm), red (~ 650 nm); operating currents typically 10–30 mA.
- **Infrared LEDs** — emit at 880–940 nm; paired with a photosensor in remote controls and intrusion alarms. Narrower viewing angle than visible LEDs, so the beam can be aimed.
- **Blinking LEDs** — contain a miniature IC that flashes the LED 1–6 times per second; need no series resistor.
- **Multicolour / RGB LEDs** — several LEDs in one package with a common anode or cathode; controlling the brightness of each mixes any colour, equal drive giving white. A back-to-back red/green pair indicates supply polarity.
- **LED displays** — seven LEDs arranged as a figure-8 (plus a decimal point) display digits; rugged, but hungrier than a liquid-crystal display.

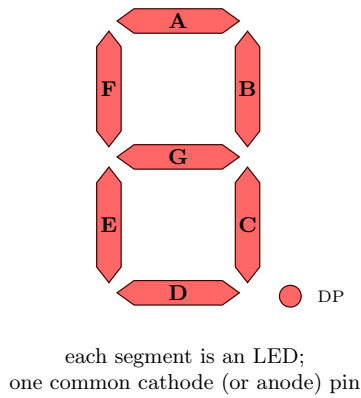


Figure 7: The seven-segment display: one LED per segment, lettered A–G plus DP, sharing a common cathode (or anode) pin.

3.3 Electrical Characteristics

An LED is a *diode*, with the same exponential I–V characteristic — but its forward voltage is set by the band gap, so every colour has its own curve. Light output is directly proportional to the forward current, and the response is fast.

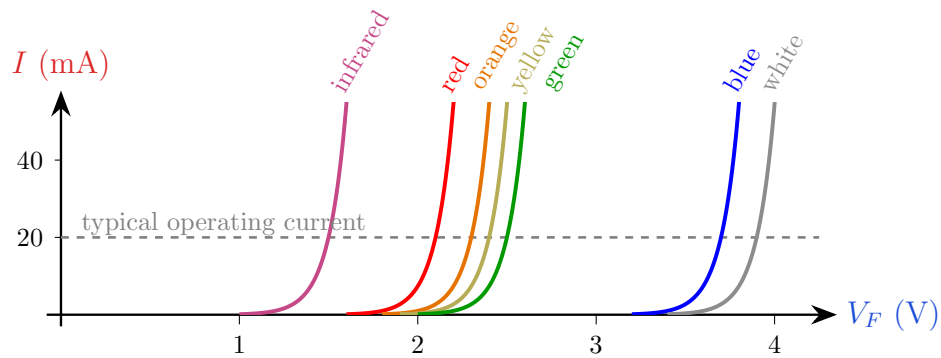


Figure 8: I–V characteristics by colour. The higher the photon energy, the further right the curve sits. The dashed line marks the typical 20 mA operating current.

Table 4: Representative forward voltages (at the usual operating current).

Wavelength	Colour	V_{LED} (V)	Dye material
940 nm	infrared	1.5	GaAlAs/GaAs
660 nm	red	1.5–1.8	GaAlAs/GaAs
635 nm	high-eff. red	2.0	GaAsP/GaP
605 nm	orange	2.1	GaAsP/GaP
585 nm	yellow	2.1	GaAsP/GaP
565 nm	green	2.1	GaP/GaP
470 nm	blue	3.6	SiC/GaN
430 nm	deep blue	3.8	SiC/GaN
white	(blue + phosphor)	3.6	SiC/GaN

One LED produces one colour. White light must be *mixed*: either a blue LED pumping a yellow phosphor (the easiest way), or red + green + blue combined (the highest-quality way).

If you don't know the recommended current of a given LED, it is usually safe to assume around 20 mA. Other ratings to respect: power dissipation (~ 100 mW typical), reverse-voltage rating, pulse current (~ 100 mA typical), and viewing angle.

3.4 Driving an LED

To light an LED, apply a voltage greater than its forward voltage V_{LED} , and *limit the current* with a series resistor to the manufacturer's recommended I_{LED} :

The series-resistor recipe

$$R_S = \frac{V_{IN} - V_{LED}}{I_{LED}}$$

KVL says it all: the supply must cover the LED's drop, and the resistor takes up the rest at the chosen current. For brightness control, add a 1 k potentiometer in series.

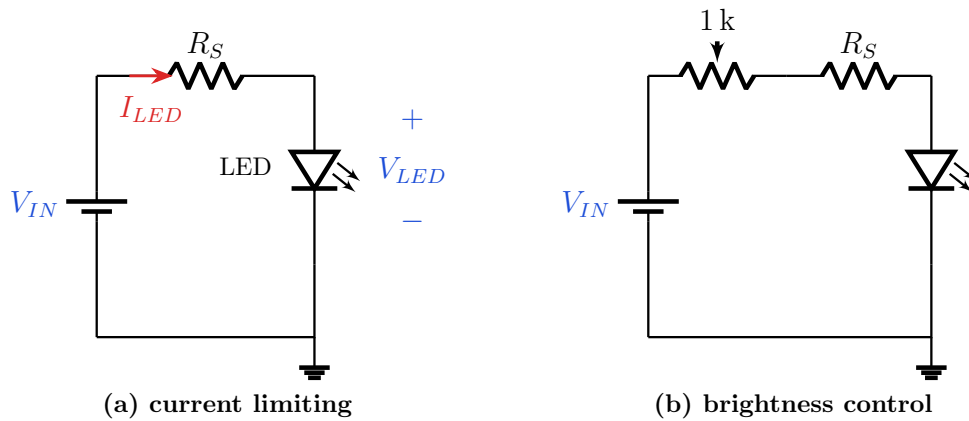


Figure 9: (a) The basic LED driver — a series resistor sets the current. (b) A potentiometer in series gives variable brightness.

Never leave the resistor out

A diode's exponential characteristic means a large voltage applied straight across a forward-biased LED produces an *enormous* current — and a dead LED. The series resistor is what turns the supply into a controlled current source.

3.4.1 Example 3 — Sizing the Series Resistor

A red LED with $V_{LED} = 1.8$ V is to run at its recommended $I_{LED} = 20$ mA from a $V_{IN} = 9$ V supply. Choose R_S , and check what power it must dissipate.

Solve: Around the loop, the supply covers the LED drop and the resistor drop:

$$V_{IN} = V_{LED} + I_{LED}R_S \quad \Rightarrow \quad 9 = 1.8 + (20 \text{ mA}) R_S$$

$$R_S = \frac{V_{IN} - V_{LED}}{I_{LED}} = \frac{9 - 1.8}{20 \text{ mA}} = \frac{7.2 \text{ V}}{20 \text{ mA}} = 360 \text{ } \Omega$$

The resistor's power:

$$P_R = I_{LED}^2 R_S = (0.020)^2 \times 360 = 0.144 \text{ W}$$

Answers

$$R_S = 360 \text{ } \Omega \quad P_R = 0.144 \text{ W} \Rightarrow \text{a quarter-watt (0.25 W) resistor is fine}$$

3.4.2 Example 4 — A Series String

Three green LEDs ($V_{LED} = 2.1$ V each, $I_{LED} = 20$ mA) are to be driven in series from a 12 V supply. Find the single series resistor required, and check the design against the 80 % guideline.

Solve: In a series string the drops simply add, so the recipe is used with the *sum* of the forward voltages:

$$R_S = \frac{V_{IN} - (V_{D1} + V_{D2} + V_{D3})}{I_{LED}} = \frac{12 - 3 \times 2.1}{20 \text{ mA}} = \frac{12 - 6.3}{0.020} = 285 \, \Omega$$

Verify: For stable, predictable current the string's total drop should not exceed about 80 % of the supply:

$$\sum V_{LED} = 6.3 \text{ V} \leq 0.8 \times 12 = 9.6 \text{ V} \quad \checkmark$$

Power in the resistor: $P = (0.020)^2 \times 285 = 0.114$ W — a quarter-watt part again.

Answers

$$R_S = 285 \, \Omega \text{ (use the standard } 300 \, \Omega) \quad \sum V_{LED} = 6.3 \text{ V } \checkmark$$

3.5 Strings, Parallels and Other Rules of Thumb

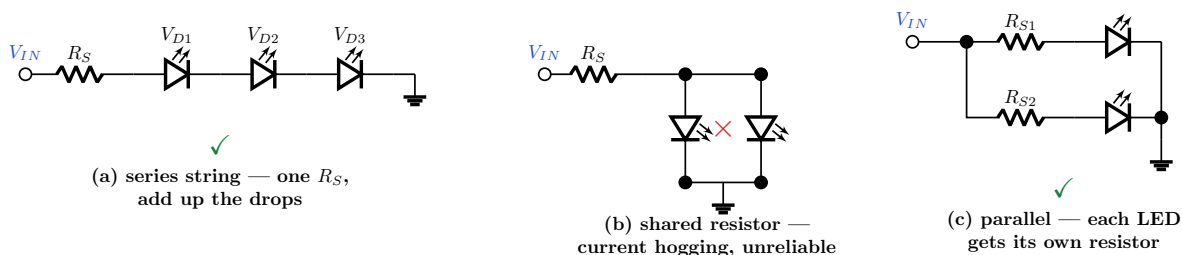


Figure 10: Series is safe with one resistor; parallel LEDs must each get their own.

- **Series** — **yes**. Add up the drops and use one resistor (Example 4).
- **Parallel with one shared resistor** — **no**. LEDs become *more* conductive as they warm up, so one LED starts hogging the current, gets warmer, hogs more ... Each parallel LED (or each series string) needs its own dropping resistor.
- **Blinking LEDs** take no series resistor at all — just stay within the recommended 3–9 V supply range.
- An LED behind a **zener diode** lights only when the input exceeds the zener's breakdown voltage — a one-component voltage-level indicator.
- Two LEDs in **reverse parallel** make a polarity indicator: one lights on each polarity, and both appear lit on AC.

one LED → one resistor	series string → sum the drops a resistor <i>each</i>	parallel →
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4 Section 25 — Photodiodes

Definition

Photodiodes are two-lead devices that convert light energy directly into electric current. Short the leads of a photodiode together and place it in the dark: no current flows. Illuminate it, and it becomes a small current source, pumping conventional current out of the anode, through the wire, and back into the cathode.

Photodiodes detect the fast pulses of near-infrared light used in remote controls and wireless links, and their very *linear* light-to-current response makes them the sensor of choice in light meters and intrusion alarms.

4.1 How a Photodiode Works

A photodiode is a *pn* junction built with a very thin *n* layer on top, so that light can reach the junction. An incoming photon with enough energy knocks a bound electron loose — creating an electron–hole pair. If the pair is created in or near the junction, the junction’s built-in field sweeps the electron to the *n* side and the hole to the *p* side. The separated charges either build up a voltage (leads open) or drive a current (leads connected).

photon absorbed → electron–hole pair → junction field separates them →
current flows

The device works with no bias at all, but it is usually operated *reverse biased*: the widened depletion region collects carriers faster, at the price of a tiny leakage in the dark.

The two operating modes

- **Photovoltaic** — no bias. The illuminated diode itself drives the current (this is exactly how a solar cell operates). Output current is nearly linear in brightness.
- **Photoconductive** — reverse biased by a battery. In the dark only the tiny **dark current** flows (nA range); light adds a much larger photocurrent. Faster response, larger output; a series resistor calibrates the reading.

Note that the photodiode is pointed *against* the battery: treat it like an ordinary conducting diode and the circuit will not work.

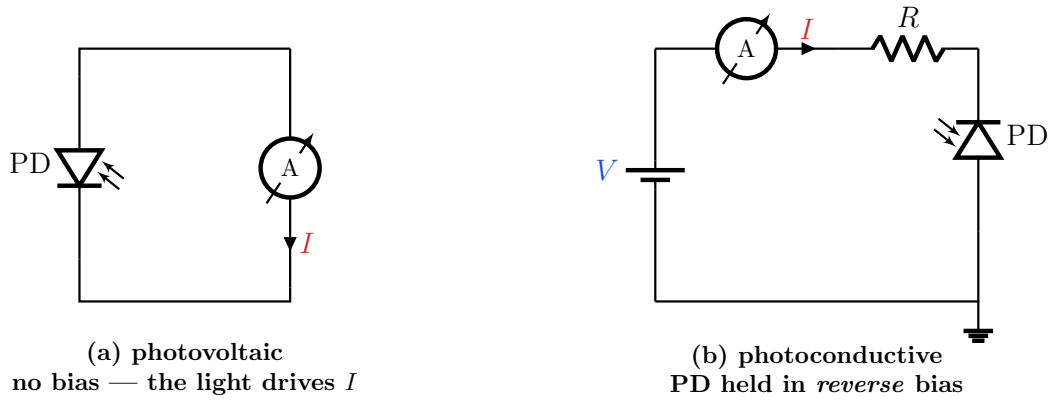


Figure 11: The two basic photodiode circuits. (a) Photovoltaic: the diode is the source. (b) Photoconductive: a battery holds the diode in reverse bias and the light modulates the reverse current.

4.2 The I–V Characteristic Under Illumination

Light adds a constant photocurrent I_{ph} flowing in the *reverse* direction, shifting the whole diode curve straight down:

$$I = I_s \left(e^{V_d/(nV_T)} - 1 \right) - I_{ph} \quad I_{ph} \propto \text{light intensity}$$

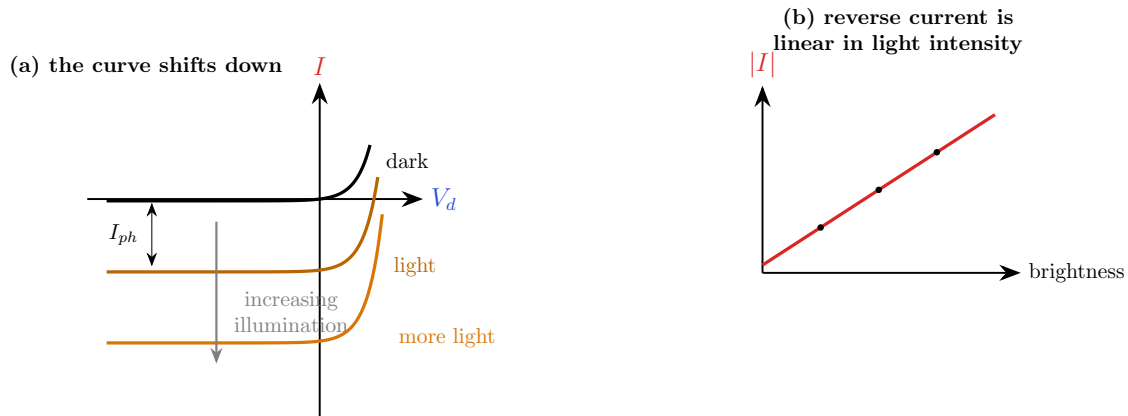


Figure 12: (a) The diode characteristic shifts down by the photocurrent I_{ph} as the illumination grows. (b) In reverse bias the measured current is essentially I_{ph} itself — linear in brightness.

more light → curve shifts further down → reverse current reads the brightness

4.2.1 Example 5 — Reading Light as a Voltage

In the photoconductive circuit of Figure 11(b), a 9 V battery reverse-biases a photodiode through $R = 100 \text{ k}\Omega$. The diode's dark current is 10 nA, and at a certain brightness its photocurrent is $I_{ph} = 2 \text{ }\mu\text{A}$. Find the voltage across R in the dark and at that brightness, and confirm the diode stays reverse biased. What happens if the brightness doubles?

Assume: The photodiode remains **reverse biased**, so the current through the loop is the diode's reverse current (dark current + photocurrent).

Solve: In the dark:

$$V_R = I_{\text{dark}} R = (10 \text{ nA})(100 \text{ k}\Omega) = 1 \text{ mV} \approx 0$$

Illuminated:

$$V_R = (I_{\text{dark}} + I_{ph}) R \approx (2 \text{ }\mu\text{A})(100 \text{ k}\Omega) = 0.2 \text{ V}$$

Verify: The diode then has $9 - 0.2 = 8.8 \text{ V}$ of reverse bias left across it — comfortably still reverse biased, so the assumption holds.

Assumption is CORRECT!

Because the response is linear, doubling the brightness doubles the photocurrent: $V_R = 0.4 \text{ V}$. The resistor converts the light level into a readable voltage — this is the front end of every light meter.

Answers

$$V_R^{\text{dark}} \approx 1 \text{ mV} \quad V_R^{\text{lit}} \approx 0.2 \text{ V} \quad V_R^{2 \times \text{lit}} \approx 0.4 \text{ V}$$

4.3 Kinds of Photodiodes

Photodiodes come in all shapes and sizes: with built-in lenses, with optical filters, optimized for speed or for sensitivity. The trade-off to remember: a *larger* surface area collects more light (more sensitivity) but responds *more slowly*.

LED and photodiode — the same junction, run in opposite directions

current in $\rightarrow$ photons out
LED (forward bias)

photons in $\rightarrow$ current out
PD (zero or reverse bias)

5 Section 26 — Optoisolators

Definition

Optoisolators (or *optocouplers*) interconnect two circuits by means of an *optical interface* — typically an LED and a phototransistor enclosed in a light-tight container. Whenever the LED side is supplied current, it emits photons; the phototransistor detects them and conducts in the second circuit. The two circuits share light, but *no electrical connection at all*.

The sensor need not be a phototransistor: LED–photodiode, LED–LASCR and lamp–photoresistor pairs all exist. (A phototransistor is simply a photodiode with transistor gain — its base current is supplied by light.)

In terms of applications, optoisolators provide **electrical isolation**: one circuit controls another without any unwanted voltages or currents sneaking across — different grounds, different supply levels, noisy versus clean sections, all safely separated.

5.1 The Three Configurations

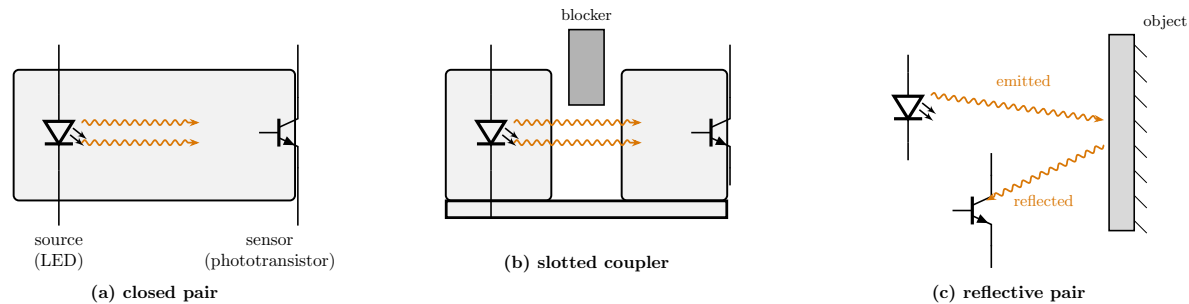


Figure 13: Optoisolator configurations. (a) The closed pair: source and sensor sealed face-to-face. (b) The slotted coupler: an open slot in the light path. (c) The reflective pair: the sensor sees the source only via reflection off an object.

Table 5: Configurations and what they are used for.

Configuration	Typical uses
Closed pair	electrical isolation, level conversion, solid-state relaying
Slotted coupler	object detection, bounce-free switching, vibration detection
Reflective pair	object detection, reflectance monitoring, tachometers, movement detection

LED side:	drive it like any LED	light crosses the gap	sensor side:
		a light-controlled switch	

5.2 Integrated Optoisolators

Closed-pair optoisolators usually come in integrated packages — the classic form is a 6-pin DIP with the LED on one side and the phototransistor on the other. Dual and quad versions put two or four isolated channels in one package.

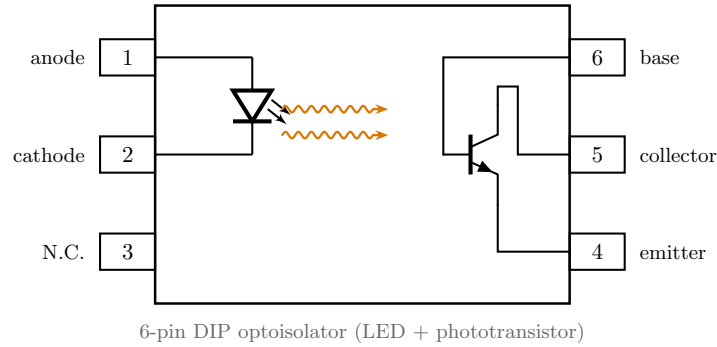


Figure 14: A 6-pin DIP optoisolator. Pins 1/2 drive the internal LED; pins 4/5 (and the rarely used base pin 6) belong to the phototransistor. Nothing but light crosses the middle.

5.2.1 Example 6 — Driving an Optoisolator

A microcontroller pin (5 V when high) must switch a load in a separate 12 V circuit through the optoisolator of Figure 14. The internal LED drops $V_{LED} = 1.2$ V and is rated for $I_{LED} = 10$ mA. Choose the LED-side series resistor, and describe the output side.

Solve: The LED side is driven exactly like any LED — with the series-resistor recipe of Section 24:

$$R_S = \frac{V_{IN} - V_{LED}}{I_{LED}} = \frac{5 - 1.2}{10 \text{ mA}} = 380 \, \Omega \quad (\text{use the standard } 390 \, \Omega)$$

On the output side, the phototransistor works as a light-controlled switch: tie its emitter to the 12 V circuit's ground and its collector through a pull-up resistor (say 10 k Ω) to +12 V. With the LED off the transistor is open and the collector sits at 12 V; with the LED on the transistor conducts and pulls the collector low.

Answers

$$R_S = 380 \, \Omega \text{ (390 } \Omega \text{ standard)}$$

The two grounds are never connected — the 5 V and 12 V circuits stay fully isolated, exchanging nothing but light.